

Thichanon Ratanasaenwan

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Education

Chiang Mai University, B.S. in Computer Science 2023 – Present

- Expected Graduation: 2027 · Major GPA: 3.54 / 4.00
- Relevant Coursework: Software Engineering, Database Systems, Computer Networks, Pattern Recognition

Technical Skills

- **Languages & Runtime:** TypeScript, JavaScript, Python, Node.js, Bun
- **Backend & APIs:** ElysiaJS, FastAPI, Express, REST API, RabbitMQ, Redis
- **AI / LLM:** RAG pipelines, LangChain, Pinecone / pgvector, OCR, multi-provider LLM integration
- **Frontend:** Next.js, React, Nuxt/Vue, Expo (React Native), Tailwind CSS
- **Database & Storage:** PostgreSQL, Prisma ORM, Drizzle ORM, MinIO / S3
- **DevOps:** Docker Compose, Caddy/Traefik, GitHub Actions, Proxmox, Coolify, Cloudflare Tunnel
- **Integrations:** LINE LIFF, LINE OA / Messaging API, Azure AD SSO

Experience

Full-Stack Developer (Freelance) — SmartMath: AI Math Learning Platform 2025 – Present

- Built a microservices RAG platform: chat over real textbooks, automated quiz generation and document summarization, using an ElysiaJS/Bun API and a FastAPI + LangChain agent with Pinecone retrieval and reranking.
- Designed an async job pipeline (RabbitMQ workers, Redis rate-limiting, SSE status) and operate the full Docker Compose deployment with PostgreSQL, MinIO and Prometheus.

Full-Stack Developer (Freelance) — Multi-Provider LLM Chat Platform 2025

- Built a ChatGPT-style chat platform with a pluggable provider layer supporting OpenAI, Anthropic and Gemini behind one abstraction, with auth and chat history.
- Architected a Next.js BFF over an async FastAPI backend with separate renderer and reranker services, shipped with Docker + Caddy.

Full-Stack Developer (Freelance) — Multi-App Repair Service Platform 2025 – Present

- Delivered four apps on one ElysiaJS/Bun API (admin, superadmin, LINE LIFF and an Expo mobile app) with role-based access, PDF generation and push notifications.
- Instrumented the backend with OpenTelemetry tracing and scheduled jobs; storage on PostgreSQL (Prisma) and S3.

Full-Stack Developer (Freelance) — Medical Conference Registration & Ordering System 2025 – Present

- Built a LINE LIFF system for event registration, QR check-in and beverage ordering: real-time queues over Socket.IO, an admin dashboard, and LINE notifications for order status.

Full-Stack Developer (Part-Time) — Membership & Rewards System 2025

- Fixed critical production bugs in a LINE-based loyalty platform used in daily store operations; implemented POS bill cancellation and promotion management, and led a UI revamp.

Full-Stack Developer (Freelance) — Nurse Phlebitis Assessment App 2025

- Converted paper-based IV-site assessment forms into a digital workflow with structured clinical-flag scoring, dashboards and LINE Login access for nursing research.
- Built automated Excel and PDF export pipelines to support nurse reporting.

Projects

Self-Hosted Infrastructure — Personal Homelab 2025 – Present

- Run my own deployment platform on Proxmox: Coolify PaaS with push-to-deploy CI/CD, exposed through a Cloudflare Tunnel with wildcard domains. This resume's portfolio site is served from it.

Security System-as-a-Service for Communities — Team Software Engineering Project (204361) 2025

- Designed the system architecture and DevOps (Docker Compose, Caddy, PostgreSQL); implemented auth, event handling and integrations; oversaw the team's GitHub workflow.